AI LESSON PLAN

Introduction to Machine Learning

A 45-minute lesson with hands-on examples, built for K-8 teachers

GRADES K-8 | 45 MIN | NO AI BACKGROUND NEEDED

Learning Objectives

- Students will define **machine learning** in their own words.
- Students will identify at least 3 everyday examples of ML in action (recommendations, voice assistants, photo tagging).
- Students will explain the difference between a **rule-based program** and a program that **learns from examples**.
- Students will participate in a hands-on sorting activity that mimics how an ML model learns.

Materials Needed

- 20-30 picture cards or small objects sorted into 2 categories (e.g., fruits vs. vegetables)
- Whiteboard or chart paper
- Access to the AI on the Streets website (optional, for the simulation demo)
- Printed copies of the Knowledge Check (last page)

Lesson Timeline

Time	Activity	Notes
0-5 min	Hook: Ask 'How does Netflix know what shows you'll like?'	Let students guess before revealing it's machine learning.
5-15 min	Direct Instruction: What is Machine Learning?	Use the 3-part definition below. Contrast with traditional 'if-then' programming.
15-30 min	Hands-On Activity: 'Teach the Computer'	See activity instructions below. Students act as the ML model.
30-40 min	Discussion + Real World Connections	Connect to the AI Traffic Simulation: the AI mode 'learns' which queue to serve.
40-45 min	Knowledge Check + Wrap-Up	Use the printable quiz on the final page.

The 3-Part Definition (write on board)

- **1. Learn:** The computer looks at lots of examples (data).
- **2. Find Patterns:** It notices what the examples have in common.
- **3. Predict:** It uses those patterns to make a guess about something new.

Activity Instructions: 'Teach the Computer'

Divide students into pairs. One student is the 'Teacher' and shows the other student (the 'Computer') 5 examples from Category A and 5 examples from Category B, without saying the category names — just showing the items. Then the Teacher shows a new, unseen item and asks the Computer to guess which category it belongs to, based only on patterns it noticed. Reveal the answer, then discuss:

- How did you decide? What pattern did you notice?
- Were you right? What made it tricky?
- This is exactly what a Machine Learning model does, except with thousands or millions of examples instead of just 10.

Connecting to the AI Traffic Simulation

Real Connection

If time allows, show Mode C (AI Adaptive) in the traffic simulation. Explain that instead of learning 'fruit vs. vegetable,' the AI traffic controller learns 'which direction has more cars waiting' and decides which light should turn green, using Reinforcement Learning, a type of machine learning where the AI gets better through trial and reward.

Differentiation Tip

For younger students, replace the sorting activity with physical objects (stuffed animals vs. toy cars). For older students, introduce the vocabulary words *training data*, *model*, and *prediction* explicitly.

PRINTABLE QUIZ

Knowledge Check

A short student quiz to wrap up the lesson

Name: ________________________________ Date: ________________

1. Machine learning is when a computer:
a) Follows the exact same rule every time
b) Learns patterns from examples and uses them to make predictions
c) Cannot be wrong

2. Give one real-world example of machine learning you use:

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3. In the 'Teach the Computer' activity, what did the Computer use to make its guess?

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4. True or False: The AI traffic light mode uses machine learning to decide which light to turn green.

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5. What is one difference between a rule-based program and a machine learning program?

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Answer Key (for teacher use): 1) b 4) True